



National  
Qualifications  
2023

**X857/76/22**

**Physics**  
**Paper 1 — Relationships sheet**

WEDNESDAY, 17 MAY

9:00 AM – 9:45 AM

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\* X 8 5 7 7 6 2 2 \*

## Relationships required for Physics Higher

$$d = \bar{v}t$$

$$s = \bar{v}t$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{1}{2}(u + v)t$$

$$F = ma$$

$$W = mg$$

$$E_w = Fd, \text{ or } W = Fd$$

$$E_p = mgh$$

$$E_k = \frac{1}{2}mv^2$$

$$P = \frac{E}{t}$$

$$p = mv$$

$$Ft = mv - mu$$

$$F = G \frac{m_1 m_2}{r^2}$$

$$t' = \frac{t}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

$$l' = l \sqrt{1 - \left(\frac{v}{c}\right)^2}$$

$$f_o = f_s \left( \frac{v}{v \pm v_s} \right)$$

$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}}$$

$$z = \frac{v}{c}$$

$$v = H_0 d$$

$$W = QV$$

$$E = mc^2$$

$$I = \frac{P}{A}$$

$$I = \frac{k}{d^2}$$

$$I_1 d_1^2 = I_2 d_2^2$$

$$E = hf$$

$$E_k = hf - hf_0$$

$$v = f\lambda$$

$$E_2 - E_1 = hf$$

$$d \sin \theta = m\lambda$$

$$n = \frac{\sin \theta_1}{\sin \theta_2}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2}$$

$$\sin \theta_c = \frac{1}{n}$$

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$$

$$I_{rms} = \frac{I_{peak}}{\sqrt{2}}$$

$$T = \frac{1}{f}$$

$$V = IR$$

$$P = IV = I^2 R = \frac{V^2}{R}$$

$$R_T = R_1 + R_2 + \dots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$V_1 = \left( \frac{R_1}{R_1 + R_2} \right) V_S$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$E = V + Ir$$

$$C = \frac{Q}{V}$$

$$Q = It$$

$$E = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2} \frac{Q^2}{C}$$

$$\text{path difference} = m\lambda \quad \text{or} \quad \left(m + \frac{1}{2}\right)\lambda \quad \text{where } m = 0, 1, 2, \dots$$

$$\text{random uncertainty} = \frac{\text{max. value} - \text{min. value}}{\text{number of values}}$$

or

$$\Delta R = \frac{R_{\max} - R_{\min}}{n}$$

## Additional relationships

### Circle

$$\text{circumference} = 2\pi r$$

$$\text{area} = \pi r^2$$

### Sphere

$$\text{area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3}\pi r^3$$

### Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Electron arrangements of elements

Group 1      Group 2  
(1)

|                                     |                                     |
|-------------------------------------|-------------------------------------|
| 1<br><b>H</b><br>1                  | 4<br><b>He</b><br>2                 |
| Hydrogen                            | (2)                                 |
| 3<br><b>Li</b><br>2,1               | 4<br><b>Be</b><br>2,2               |
| Lithium                             | Beryllium                           |
| 11<br><b>Na</b><br>2,8,1            | 12<br><b>Mg</b><br>2,8,2            |
| Sodium                              | Magnesium                           |
| 19<br><b>K</b><br>2,8,8,1           | 20<br><b>Ca</b><br>2,8,8,2          |
| Potassium                           | Calcium                             |
| 37<br><b>Rb</b><br>2,8,18,8,1       | 38<br><b>Sr</b><br>2,8,18,8,2       |
| Rubidium                            | Strontium                           |
| 55<br><b>Cs</b><br>2,8,18,18,8,1    | 56<br><b>Ba</b><br>2,8,18,18,8,2    |
| Caesium                             | Barium                              |
| 87<br><b>Fr</b><br>2,8,18,32,18,8,1 | 88<br><b>Ra</b><br>2,8,18,32,18,8,2 |
| Francium                            | Radium                              |

Key

|                      |
|----------------------|
| Atomic number        |
| Symbol               |
| Electron arrangement |
| Name                 |

Transition elements

|                                     |                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |                                       |
|-------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (3)                                 | (4)                                   | (5)                                   | (6)                                   | (7)                                   | (8)                                   | (9)                                   | (10)                                  | (11)                                  | (12)                                  |
| 21<br><b>Sc</b><br>2,8,9,2          | 22<br><b>Ti</b><br>2,8,10,2           | 23<br><b>V</b><br>2,8,11,2            | 24<br><b>Cr</b><br>2,8,13,1           | 25<br><b>Mn</b><br>2,8,13,2           | 26<br><b>Fe</b><br>2,8,14,2           | 27<br><b>Co</b><br>2,8,15,2           | 28<br><b>Ni</b><br>2,8,16,2           | 29<br><b>Cu</b><br>2,8,18,1           | 30<br><b>Zn</b><br>2,8,18,2           |
| Scandium                            | Titanium                              | Vanadium                              | Chromium                              | Manganese                             | Iron                                  | Cobalt                                | Nickel                                | Copper                                | Zinc                                  |
| 39<br><b>Y</b><br>2,8,18,9,2        | 40<br><b>Zr</b><br>2,8,18,10,2        | 41<br><b>Nb</b><br>2,8,18,12,1        | 42<br><b>Mo</b><br>2,8,18,13,1        | 43<br><b>Tc</b><br>2,8,18,13,2        | 44<br><b>Ru</b><br>2,8,18,15,1        | 45<br><b>Rh</b><br>2,8,18,16,1        | 46<br><b>Pd</b><br>2,8,18,18,0        | 47<br><b>Ag</b><br>2,8,18,18,1        | 48<br><b>Cd</b><br>2,8,18,18,2        |
| Yttrium                             | Zirconium                             | Niobium                               | Molybdenum                            | Technetium                            | Ruthenium                             | Rhodium                               | Palladium                             | Silver                                | Cadmium                               |
| 57<br><b>La</b><br>2,8,18,18,9,2    | 72<br><b>Hf</b><br>2,8,18,32,10,2     | 73<br><b>Ta</b><br>2,8,18,32,11,2     | 74<br><b>W</b><br>2,8,18,32,12,2      | 75<br><b>Re</b><br>2,8,18,32,13,2     | 76<br><b>Os</b><br>2,8,18,32,14,2     | 77<br><b>Ir</b><br>2,8,18,32,15,2     | 78<br><b>Pt</b><br>2,8,18,32,17,1     | 79<br><b>Au</b><br>2,8,18,32,18,1     | 80<br><b>Hg</b><br>2,8,18,32,18,2     |
| Lanthanum                           | Hafnium                               | Tantalum                              | Tungsten                              | Rhenium                               | Osmium                                | Iridium                               | Platinum                              | Gold                                  | Mercury                               |
| 89<br><b>Ac</b><br>2,8,18,32,18,9,2 | 104<br><b>Rf</b><br>2,8,18,32,32,10,2 | 105<br><b>Db</b><br>2,8,18,32,32,11,2 | 106<br><b>Sg</b><br>2,8,18,32,32,12,2 | 107<br><b>Bh</b><br>2,8,18,32,32,13,2 | 108<br><b>Hs</b><br>2,8,18,32,32,14,2 | 109<br><b>Mt</b><br>2,8,18,32,32,15,2 | 110<br><b>Ds</b><br>2,8,18,32,32,17,1 | 111<br><b>Rg</b><br>2,8,18,32,32,18,1 | 112<br><b>Cn</b><br>2,8,18,32,32,18,2 |
| Actinium                            | Rutherfordium                         | Dubnium                               | Seaborgium                            | Bohrium                               | Hassium                               | Meitnerium                            | Darmstadtium                          | Roentgenium                           | Copernicium                           |

|                                   |                                   |                                   |                                   |                                   |                                   |
|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| 2<br><b>He</b><br>2               | 6<br><b>C</b><br>2,4              | 7<br><b>N</b><br>2,5              | 8<br><b>O</b><br>2,6              | 9<br><b>F</b><br>2,7              | 10<br><b>Ne</b><br>2,8            |
| Helium                            | Carbon                            | Nitrogen                          | Oxygen                            | Fluorine                          | Neon                              |
| (13)                              | (14)                              | (15)                              | (16)                              | (17)                              | (18)                              |
| 5<br><b>B</b><br>2,3              | 6<br><b>C</b><br>2,4              | 7<br><b>N</b><br>2,5              | 8<br><b>O</b><br>2,6              | 9<br><b>F</b><br>2,7              | 10<br><b>Ne</b><br>2,8            |
| Boron                             | Carbon                            | Nitrogen                          | Oxygen                            | Fluorine                          | Neon                              |
| 13<br><b>Al</b><br>2,8,3          | 14<br><b>Si</b><br>2,8,4          | 15<br><b>P</b><br>2,8,5           | 16<br><b>S</b><br>2,8,6           | 17<br><b>Cl</b><br>2,8,7          | 18<br><b>Ar</b><br>2,8,8          |
| Aluminium                         | Silicon                           | Phosphorus                        | Sulfur                            | Chlorine                          | Argon                             |
| 31<br><b>Ga</b><br>2,8,18,3       | 32<br><b>Ge</b><br>2,8,18,4       | 33<br><b>As</b><br>2,8,18,5       | 34<br><b>Se</b><br>2,8,18,6       | 35<br><b>Br</b><br>2,8,18,7       | 36<br><b>Kr</b><br>2,8,18,8       |
| Gallium                           | Germanium                         | Arsenic                           | Selenium                          | Bromine                           | Krypton                           |
| 49<br><b>In</b><br>2,8,18,18,3    | 50<br><b>Sn</b><br>2,8,18,18,4    | 51<br><b>Sb</b><br>2,8,18,18,5    | 52<br><b>Te</b><br>2,8,18,18,6    | 53<br><b>I</b><br>2,8,18,18,7     | 54<br><b>Xe</b><br>2,8,18,18,8    |
| Indium                            | Tin                               | Antimony                          | Tellurium                         | Iodine                            | Xenon                             |
| 81<br><b>Tl</b><br>2,8,18,32,18,3 | 82<br><b>Pb</b><br>2,8,18,32,18,4 | 83<br><b>Bi</b><br>2,8,18,32,18,5 | 84<br><b>Po</b><br>2,8,18,32,18,6 | 85<br><b>At</b><br>2,8,18,32,18,7 | 86<br><b>Rn</b><br>2,8,18,32,18,8 |
| Thallium                          | Lead                              | Bismuth                           | Polonium                          | Astatine                          | Radon                             |

Group 3      Group 4      Group 5      Group 6      Group 7      Group 8  
(18)

Lanthanides

|                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |
|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| 57<br><b>La</b><br>2,8,18,18,9,2 | 58<br><b>Ce</b><br>2,8,18,20,8,2 | 59<br><b>Pr</b><br>2,8,18,21,8,2 | 60<br><b>Nd</b><br>2,8,18,22,8,2 | 61<br><b>Pm</b><br>2,8,18,23,8,2 | 62<br><b>Sm</b><br>2,8,18,24,8,2 | 63<br><b>Eu</b><br>2,8,18,25,8,2 | 64<br><b>Gd</b><br>2,8,18,25,9,2 | 65<br><b>Tb</b><br>2,8,18,27,8,2 | 66<br><b>Dy</b><br>2,8,18,28,8,2 | 67<br><b>Ho</b><br>2,8,18,29,8,2 | 68<br><b>Er</b><br>2,8,18,30,8,2 | 69<br><b>Tm</b><br>2,8,18,31,8,2 | 70<br><b>Yb</b><br>2,8,18,32,8,2 | 71<br><b>Lu</b><br>2,8,18,32,9,2 |
| Lanthanum                        | Cerium                           | Praseodymium                     | Neodymium                        | Promethium                       | Samarium                         | Europium                         | Gadolinium                       | Terbium                          | Dysprosium                       | Holmium                          | Erbium                           | Thulium                          | Ytterbium                        | Lutetium                         |

Actinides

|                                     |                                      |                                     |                                    |                                     |                                     |                                     |                                     |                                     |                                     |                                     |                                      |                                      |                                      |                                      |
|-------------------------------------|--------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| 89<br><b>Ac</b><br>2,8,18,32,18,9,2 | 90<br><b>Th</b><br>2,8,18,32,18,10,2 | 91<br><b>Pa</b><br>2,8,18,32,20,9,2 | 92<br><b>U</b><br>2,8,18,32,21,9,2 | 93<br><b>Np</b><br>2,8,18,32,22,9,2 | 94<br><b>Pu</b><br>2,8,18,32,24,8,2 | 95<br><b>Am</b><br>2,8,18,32,25,9,2 | 96<br><b>Cm</b><br>2,8,18,32,25,9,2 | 97<br><b>Bk</b><br>2,8,18,32,27,8,2 | 98<br><b>Cf</b><br>2,8,18,32,28,8,2 | 99<br><b>Es</b><br>2,8,18,32,29,8,2 | 100<br><b>Fm</b><br>2,8,18,32,30,8,2 | 101<br><b>Md</b><br>2,8,18,32,31,8,2 | 102<br><b>No</b><br>2,8,18,32,32,8,2 | 103<br><b>Lr</b><br>2,8,18,32,32,9,2 |
| Actinium                            | Thorium                              | Protactinium                        | Uranium                            | Neptunium                           | Plutonium                           | Americium                           | Curium                              | Berkelium                           | Californium                         | Einsteinium                         | Fermium                              | Mendelevium                          | Nobelium                             | Lawrencium                           |